WAEP Semester One Examination, 2015

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

SOLUTIONS

Student number:	In figures	
	In words	________________
	Your name	________________

Time allowed for this section

Reading time before commencing work: five minutes
Working time for section: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of exam
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Additional working space pages at the end of this Question/Answer booklet are for planning or continuing an answer. If you use these pages, indicate at the original answer, the page number it is planned/continued on and write the question number being planned/continued on the additional working space page.

5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you do not use pencil, except in diagrams.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free **35% (52 Marks)**

This section has **seven (7)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time for this section is 50 minutes.

Question 1 **(5 marks)**

Solve the following equations for x.

(a) $\dfrac{3x}{2}=\dfrac{6}{5}$. (1 mark)

$$3x=\frac{12}{5}$$
$$x=\tfrac{4}{5}$$

(b) $\dfrac{2x+3}{5}-\dfrac{x-2}{4}=2$. (2 marks)

$$4(2x+3)-5(x-2)=40$$
$$8x+12-5x+10=40$$
$$3x=18$$
$$x=6$$

(c) $4x^2=6x$. (2 marks)

$$4x^2-6x=0$$
$$2x(2x-3)=0$$
$$x=0,\ x=1.5$$

Question 2 **(10 marks)**

(a)

(i) What are the coordinates of the y-intercept of the graph of $y=(x-2)(x+3)$? (1 mark)

(0, -6)

(ii) State the number of roots of the graph of $y=2x^2+1$. (1 mark)

None

(iii) What are the coordinates of the turning point of the graph of $y=3(x+4)^2+2$? (1 mark)

(-4, 2)

(iv) What is the equation of the line of symmetry of the graph of $y=(x-3)(x-7)$? (1 mark)

$x=5$

(b) Determine the equation of the quadratic function that passes through the point (0, -9) and has a turning point at (-2, 3). (2 marks)

$$-9=a(0+2)^2+3 \Rightarrow a=-3$$

$$y=-3(x+2)^2+3$$

$$\left(=-3x^2-12x-9\right)$$

(c) Solve the following equations.

(i) $x^2 + 4x + 6 = 2x^2 + 5x - 6$. (2 marks)

$$
\boxed{\begin{aligned} 0 &= x^2 + x - 12 \\ (x+4)(x-3) &= 0 \\ x &= -4,\ x = 3 \end{aligned}}
$$

(ii) $2(x-2)^2 = 100$. (2 marks)

$$
\boxed{\begin{aligned} (x-2)^2 &= 50 \\ x - 2 &= \pm 5\sqrt{2} \\ x &= 2 + 5\sqrt{2},\ x = 2 - 5\sqrt{2} \end{aligned}}
$$

Question 3 **(7 marks)**

(a) A line has equation $3x-2y=12$. Determine

(i) the gradient of the line. (1 mark)

$$2y=3x-12 \;\Rightarrow\; m=\tfrac{3}{2}$$

(ii) the coordinates of the x-intercept of the line. (1 mark)

(4, 0)

(iii) the coordinates of the y-intercept of the line. (1 mark)

(0, -6)

(b) Determine the equation of the line that is perpendicular to the line in (a) and that passes through the point (6, -1), giving your answer in the form $ax+by=c$, where a,b and c are integers. (2 marks)

$$m_{\perp}=-\tfrac{2}{3}$$

$$y-(-1)=-\tfrac{2}{3}(x-6)$$

$$3y+3=-2x+12$$
$$3y+2x=9$$

(c) The mid-point of $A(-4,7)$ and B is $C(2,-2)$. Determine the coordinates of the midpoint of B and C. (2 marks)

$$B(2+(2-(-4)),-2+(-2-7))=B(8,-11)$$

$$\left(\frac{2+8}{2},\frac{-2-11}{2}\right)=(5,-6.5)$$

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 4 **(9 marks)**

(a) State the degree of the polynomial $x(x^2+1)(x^2-2)$. (1 mark)

Degree 5

(b) Determine the natural domain and corresponding range for the function $h(x)=\sqrt{9-x}$. (2 marks)

Domain:
$9-x\geq 0 \;\Rightarrow\; \{x : x\in \mathbb{R},\ x\leq 9\}$

Range:
$\{y : y\in \mathbb{R},\ y\geq 0\}$

(c) Comment on the behaviour of the following graphs as $x\to\infty$.

(i) $y=1-x^7$. (1 mark)

$y\to -\infty$

(ii) $y=1-\dfrac{1}{x}$. (1 mark)

$y\to 1$

(d) Let $f(x)=x^3+4x^2-7x-10$ over the domain $\{x : x=-1,\ 0,\ 1,\ 2\}$.

(i) Determine the range of $f(x)$. (2 marks)

$f(-1)=0$
$f(0)=-10$
$f(1)=-12$
$f(2)=0$
$\{y : y=-12,\ -10,\ 0\}$

(ii) Factorise $x^3+4x^2-7x-10$. (2 marks)

$$\begin{aligned} x^3+4x^2-7x-10 &= (x-2)(x^2+6x+5)\\ &= (x-2)(x+1)(x+5) \end{aligned}$$

Question 5 **(8 marks)**

(a) Simplify $\dfrac{11!}{6! \times 7!}$. (2 marks)

$$\frac{11\times10\times9\times8\times7!}{6\times5\times4\times3\times2\times7!}=\frac{11\times10\times9}{6\times5\times3}$$
$$=11$$

(b) The sum of all the numbers in one row of Pascal's triangle is 16.

(i) Write down all the numbers in this row of Pascal's triangle. (1 mark)

1, 4, 6, 4, 1

(ii) Expand $(x-3)^4$. (3 marks)

$$(x-3)^4 = x^4 + 4x^3(-3) + 6x^2(-3)^2 + 4x(-3)^3 + (-3)^4$$
$$= x^4 - 12x^3 + 54x^2 - 108x + 81$$

(c) Factorise $x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 - 6x + 1$. (2 marks)

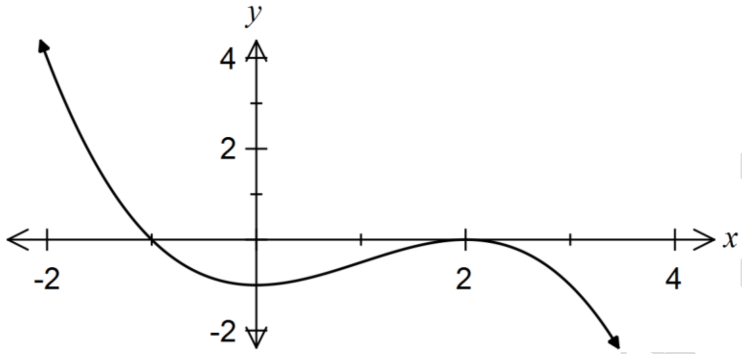
$$1(-1)^0 x^6 + 6(-1)^1 x^5 + 15(-1)^2 x^4 + 20(-1)^3 x^3 + 15(-1)^4 x^2 + 6(-1)^5 x^1 + 1(-1)^6$$

$$= (x-1)^6$$

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 6 **(8 marks)**

(a) The graph below shows the function $y = k(x-a)(x-b)(x-c)$, where a, b, c and k are real constants.



Determine the value of k. (2 marks)

Using roots of -1 and 2 and y-intercept of -1:

$y = k(x+1)(x-2)(x-2)$

$-1 = k(1)(-2)(-2) \Rightarrow k = -\frac{1}{4}$

(b) The function g is given by $g(x) = 2(x-1)(x+1)(2x+3)$.

(i) Determine the coordinates of all axes intercepts of the graph of $y = g(x)$. (2 marks)

y-intercept: (0, -6)

$x-$intercepts: (1, 0), (-1,0) and (-1.5, 0)

(ii) $g(x)$ can also be expressed in the form $ax^3 + bx^2 + cx + d$. Determine the value of the coefficient b. (2 marks)

$$2(x-1)(x+1)(2x+3) = 2(x^2-1)(2x+3)$$
$$= 4x^3 + 6x^2 - 4x - 3 \Rightarrow b = 6$$

(c) Solve $(3x-4)(x^2-7x-18) = 0$. (2 marks)

$$(3x-4)(x^2-7x-18) = 0$$
$$(3x-4)(x+2)(x-9) = 0 \Rightarrow x = \frac{4}{3},\ x = -2,\ x = 9$$

Question 7 **(5 marks)**

(a) Solve $2\cos 2x - \sqrt{3} = 0$ for $0 \le x \le \pi$. (2 marks)

$$\cos 2x = \frac{\sqrt{3}}{2}$$
$$2x = \frac{\pi}{6}, \frac{11\pi}{6}$$
$$x = \frac{\pi}{12}, \frac{11\pi}{12}$$

(b) Solve $\sin x \cos 10° - \cos x \sin 10° = 0.5$ for $0 \le x \le 360°$. (3 marks)

$$\sin x \cos 10° - \cos x \sin 10° = 0.5$$
$$\sin(x - 10°) = 0.5$$
$$x - 10° = 30°,\ 150°$$
$$x = 40°,\ 160°$$

End of questions

Additional working space

Question number: _________